

ZË-R0 v0.2 Note – Morphological Masking & Recovery in Albanian200 (Controlled Ablation)

What this is

This note reports a **diagnostic** result from ZË-R0's deterministic analysis harnesses.

It tests whether **schwa-heavy morphology** (notably për-/përë-) can mask root-aligned carrier primaries in a small, root-anchored Albanian subset.

This is **not** a claim about “semantic proof”, and it is **not** a claim about real dialect pronunciation.

The “Gegë” condition below is a **controlled perturbation** (simulation) used as an ablation probe.

Definitions (what is being measured)

- **carrierP(word)**: the **primary** vowel carrier extracted from IPA by the SSOT phonetic extractor.
- **anchorP(root)**: the primary vowel carrier extracted from the anchor root's IPA.
- **retains**: `carrierP(compound) == anchorP(anchor)`.

Inputs

- Root-anchored stress set (N=28):
 - `tests/research/albanian200.compoundStress.v0.2.txt``
- “Gegë-sim” probe set (N=28, row-aligned):
 - `tests/research/albanian200.gegProbe.v0.1.txt``

Harness outputs (baseline-locked artifacts)

- Compound Stress (root retention):
 - `tests/validation/baselines/compoundStress.albanian200.v0.2.md``
- Gegë Probe (ablation via simulated schwa-drop):
 - `tests/validation/baselines/gegProbe.albanian200.v0.1.md``

Results (verbatim from harness reports)

Compound Stress v0.2 – root-anchored retention

Retention summary (`carrierP(compound) == carrierP(anchor)`), N=14 per tag:

- **position**: 64.3% (9/14), $p=0.013$
- **order**: 100.0% (14/14), $p=0.001$

Mismatch cases in ****position**** (all are për- prefixed; compound carrierP becomes Ě while anchor is A/E):

- përballë: Ě vs A
- përtej: Ě vs E
- përreth: Ě vs E
- përparim: Ě vs A
- përparësi: Ě vs A

Gegë Probe v0.1 – Ě-drop diagnostic (simulated)

Retention by tag (Tosk-anchored vs Gegë-sim), N=14 per tag:

- ****order****: 100.0% → 100.0% (14/14, p=0.001 → 14/14, p<0.001)
- ****position****: 64.3% → 100.0% (9/14, p=0.010 → 14/14, p=0.001)

Delta cases (recoveries under simulated schwa-drop):

- përballë: Ě/A → A/A
- përtej: Ě/E → E/E
- përreth: Ě/E → E/E
- përparim: Ě/A → A/A
- përparësi: Ě/A → A/A

Interpretation (what this supports – and what it does NOT)

Supported by the diagnostic:

- In this anchored subset, the ****për- prefix schwa**** can ****dominate carrierP**** under standard/Tosk IPA, producing ****apparent washout**** relative to the anchor root.
- Under a controlled schwa-drop perturbation, the masked cases ****recover**** to anchor-aligned carrier primaries.

Not supported / not claimed here:

- No claim that this is “true Gegë pronunciation”.
- No claim of a universal sound-meaning law.
- No claim that Albanian “proves” non-arbitrariness.

This is best read as a ****mechanistic masking-and-recovery diagnostic****: the harness can detect when a consistent surface vowel (Ě) is acting as a shield over a root-aligned carrier primary (A/E) in prefixed forms.

Reproduce (exact commands)

```bash

```
npm install
npm run gate:quick
```

```
npm run research:albanian200:compoundStress:v0.2
npm run research:albanian200:gegProbe
\\
```